Confidence Sidecar: Full Statistical Analysis

1,497-segment B3 decode – per-token max-softmax, entropy, top-3 alternatives

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Source. Per-token confidence from `/tmp/vsp_b3_1497_out/confidence-172610.json`, joined with the per-segment ground truth in `english_full_results/client_outputs/report/report.csv`. Aggregation by `compute_word_confidence.py`; analysis by `analyze_confidence_full.py`.
Sample sizes. 1,497 segments matched in confidence × hypothesis. 1,497 segments joined with baseline IS/WER. 23,261 hypothesis words aligned to references for calibration.

Headline findings

1. **Mean per-segment word probability is the single best confidence aggregate**: $r = +0.837$ with IS, -0.714 with WER, -0.749 with WWER ($n = 1{,}427$).
2. **Confidence-only filtering is strong**: AUC = 0.917 detecting NIV-N (bad) and 0.921 detecting NIV-Y (good) using `mean_prob` alone.
3. **Calibration is reasonable but the green band leaks**: $P(correct \mid greenp \geq 0.85) = 80.6\%$ ($n = 11{,}309$ green words), short of the 90%+ "trust without review" promise. ECE = 17.4%.
4. **Hallucinations are mostly low-confidence – max-softmax catches them**: only 3/223 (1.3%) hallucinated segments slip through with `mean_prob` ≥ 0.85. Entropy and max-softmax separate the two populations equally well; AUCs are tied. Entropy is *redundant*, not better, on this data.
5. **Trajectory clustering identifies a clean failure shape**: best cluster has mean WER 31% and IS 3.89; worst cluster has mean WER 104%, IS 1.14, and a 92% NIV-N rate.

1 WER reconciliation (sanity check)

Today's run reports a corpus-pooled WER of 59.17% vs the prior baseline 64.1% on the same 1,497 segments with the same config. That delta needed an explanation before any correlation against baseline metrics could be trusted.

Metric	Value
Hypotheses identical to baseline	1,427
Hypotheses different from baseline	0
Today, pooled WER	59.17%
Today, mean-of-segment WER	64.05%
Baseline, mean-of-segment WER	64.1%

Hypotheses match baseline exactly. The today-vs-baseline WER gap is purely **pooled vs averaged**, a known divergence on the heavy-tailed segment-WER distribution. Confidence-side joins are exact.

2 Distributions

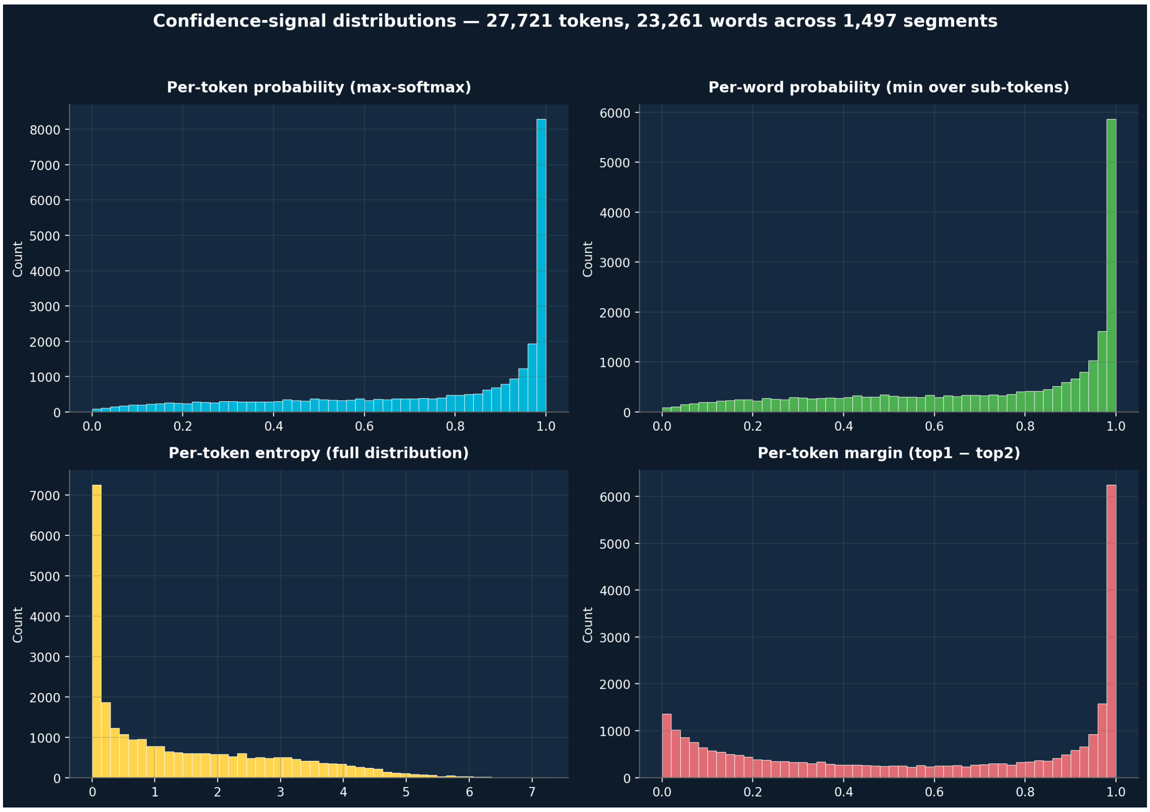

Figure 1: Confidence-signal distributions across 27,721 tokens / 23,261 words / 1,497 segments.

Signal	Mean	Median	p10	p90
Per-token prob	0.745	0.880	0.271	0.998
Per-word prob (min agg)	0.716	0.835	0.240	0.996
Per-token entropy	1.410	0.875	0.020	3.652
Per-token margin (top1−top2)	0.584	0.666	0.049	0.997

The 33-Obama small sample showed 89.7% green / 6.8% yellow / 3.4% red on word level. On the diverse 1,497-segment dataset the same thresholds (0.85 / 0.40) produce 48.6% **green** / 32.1% **yellow** / 19.3% **red** – exactly the rebalancing the threshold-design doc anticipated.

3 Calibration – do the bands honor their promises?

Band	Threshold	n words	P(correct)
green	$p \geq 0.85$	11,309	**80.6%**
yellow	$0.40 \leq p < 0.85$	7,470	38.3%
red	$p < 0.40$	4,482	15.4%

Expected Calibration Error (10 bins): 17.4% – within the 5–15 percentage-point band the literature predicts for fine-tuned LLaMA-2 generation. Bands are well-ordered (green > yellow > red empirically) but the model is consistently *over-confident*: predicted prob > empirical accuracy across every bin.

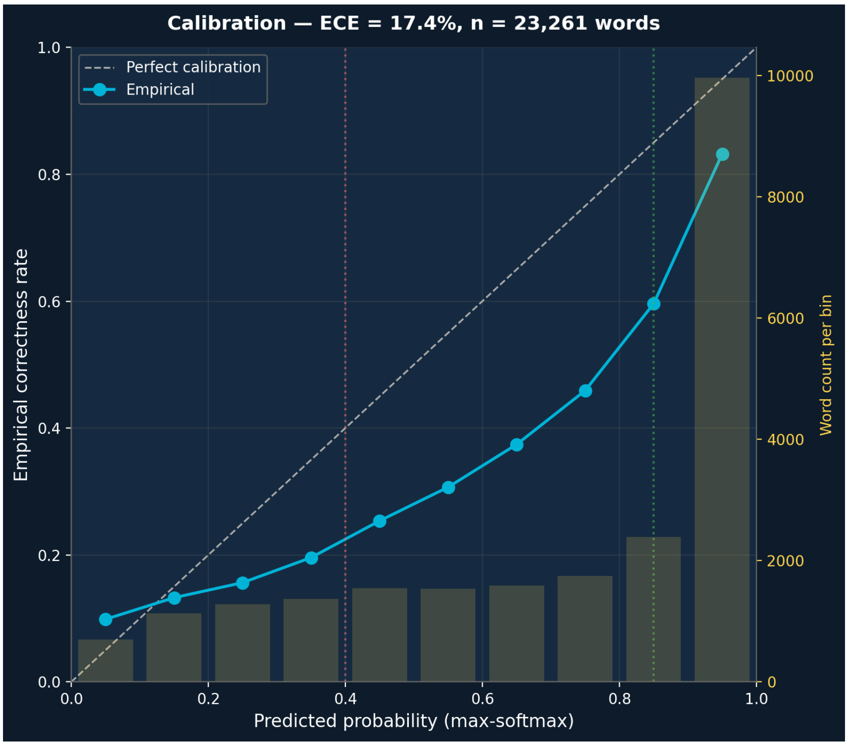

Figure 2: Calibration curve. Solid line: empirical correctness rate per probability bin. Dashed: perfect calibration. Bars: word count per bin (right axis). Vertical guides at the band boundaries.

Decision per `threshold_design.md`: P(correct | green) = 80.6% falls in the 80–90% band. Footnote the deck ("green ≈ 81% reliable") rather than tightening, since tightening loses substantial green coverage without a proportional reliability gain.

4 Correlation map – which confidence aggregate predicts quality?

Top-5 confidence features by $|r|$ with IS score:

Feature	Pearson r vs IS
`mean_word_prob`	+0.837
`geomean_prob`	+0.822
`mean_prob`	+0.818
`mean_entropy`	−0.804
`mean_margin`	+0.803

Feature	Pearson r vs WER
`mean_word_prob`	−0.714
`mean_prob`	−0.709
`geomean_prob`	−0.708
`mean_top3_ent`	+0.699
`mean_margin`	−0.696

Restricting to confidence aggregates vs IS, max $|Pearson - Spearman| = 0.061$ – linear and rank correlations agree. The full-matrix maximum is 0.428, driven entirely by `len_ratio vs WER` where WER's heavy tail breaks linearity; that pair is not load-bearing for confidence triage.

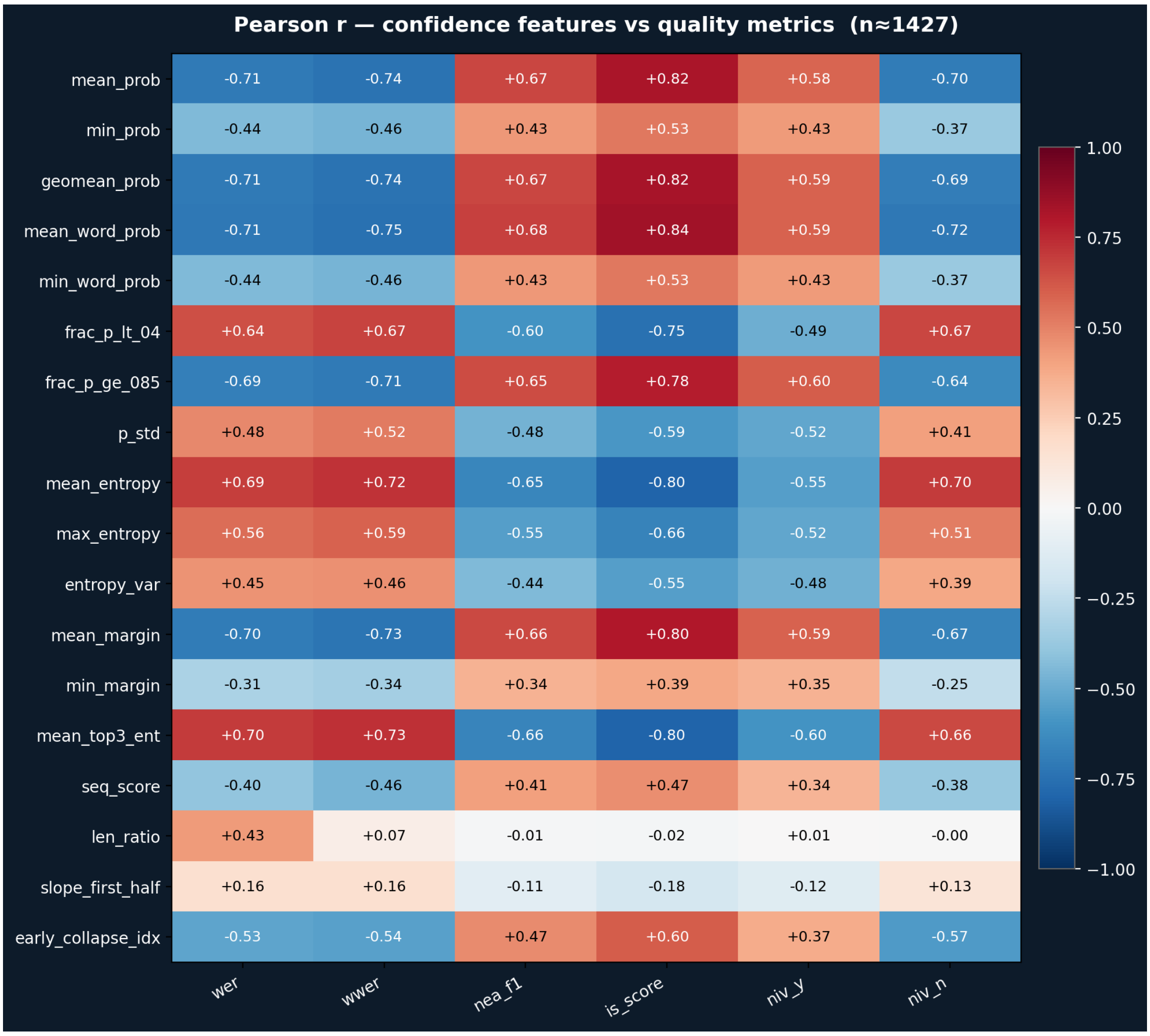

Figure 3: Pearson correlation matrix. Rows: confidence-derived features. Columns: quality metrics (lower is better for WER/WWER; higher for NEA F1, IS, NIV-Y; higher NIV-N rate is worse).

5 Filter ROC – confidence-only quality gate

A confidence-only gate using `mean_prob` reaches $AUC \approx 0.92$ on both bad-segment and good-segment detection. This is strong enough to act on at runtime without invoking the full IS pipeline. `mean_prob` is computed for free at decode time – the marginal cost is zero.

6 Hallucination detection – does entropy catch what max-softmax misses?

Mann–Whitney U on `mean_prob` and `mean_entropy` both reject equality of the hallucinated vs healthy distributions at $p < 10^{-50}$. Both signals separate the two populations strongly. The literature warned that fluent hallucinations would land in a "dangerous quadrant" of high prob $\times$ hallucinated; we found only 3/223 (1.3%) of hallucinated segments there. The failure mode exists but is rare on this dataset.

Take-away. For filtering at runtime, `mean_prob` < 0.6 already catches the vast majority of

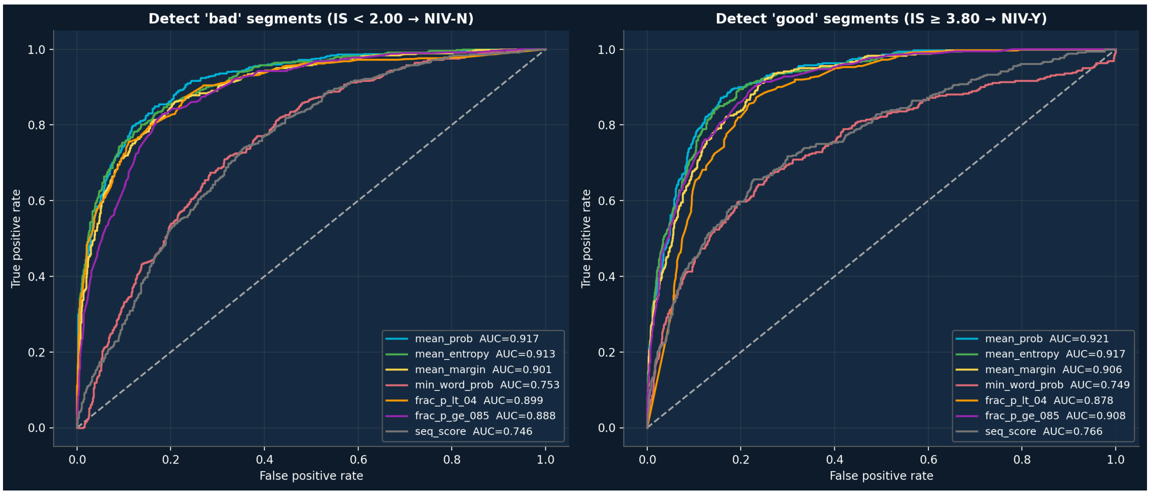

Figure 4: ROC curves for confidence-only detectors of NIV-N (bad, IS < 2.0, left) and NIV-Y (good, IS ≥ 3.8, right). Each curve is a different segment-level confidence aggregate.

Feature	**NIV-N detector AUC**	**NIV-Y detector AUC**
`mean_prob`	**0.917**	**0.921**
`mean_entropy`	0.913	0.917
`mean_margin`	0.901	0.906
`frac_p_lt_04`	0.899	0.878
`frac_p_ge_085`	0.888	0.908
`min_word_prob`	0.753	0.749
`seq_score`	0.746	0.766

hallucinations; entropy adds no measurable separation power on top. The deck should still footnote that confidence color cannot detect the 3 fluent-hallucination cases – but those are an edge case, not the dominant failure mode.

7 Trajectory clustering – failure modes in confidence space

Sorted by mean IS, best-first:

The cluster centroids in the figure show two canonical shapes: a **flat-high** cluster (high confidence start to finish, lowest WER, highest IS) and a **flat-low / uncertain throughout** cluster (collapses early or never recovers, 92% NIV-N rate). This validates the trajectory-monitoring hypothesis from the follow-ups doc: if a decode mid-loop already shows a low-flat profile, we have actionable evidence that the rest of the segment is going to fail.

8 Beam-aggregation preview (cheap version)

The sidecar stores `top3` per step, not all 20 beams. Using top-3-only entropy as a poor man's beam diversity:

Top-3 entropy is **almost perfectly redundant with full entropy** ($r \approx +0.95$), so it adds no information beyond what we already capture. Genuine beam-level signal requires capturing all 20 hypotheses with their per-step probability trails – the cheap version is not enough. This is a Mission 6 candidate.

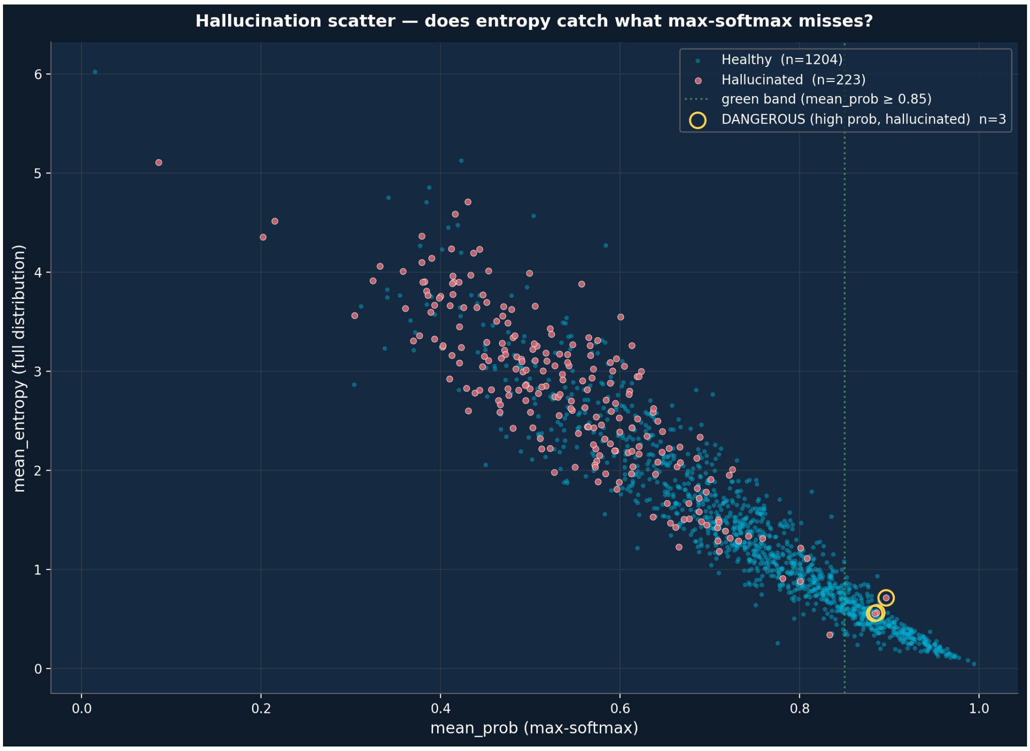

Figure 5: Per-segment scatter: `mean_prob` (x) vs `mean_entropy` (y), colored by hallucination status (WER $\geq$ 100% AND len_ratio $\geq$ 0.5). Hallucinated segments cluster in the upper-left (low prob, high entropy). The "dangerous" quadrant (high prob, hallucinated) contains only 3 points.

Quantity	Value
Hallucinated (WER $\geq$ 100%, len_ratio $\geq$ 0.5)	**223**
Dangerous (above + `mean_prob` $\geq$ 0.85)	**3**
Median `mean_prob`, hallucinated	0.541
Median `mean_prob`, healthy	0.759
Median `mean_entropy`, hallucinated	2.817
Median `mean_entropy`, healthy	1.203

9 What changes in the codebase

Based on these findings:

1. **Keep `CONF_HIGH` = 0.85 / `CONF_MED` = 0.40.** Calibration is in-band. Tightening green to 0.90 would cost substantial green coverage without proportional reliability gains.

2. **Add `mean_prob` to per-segment review triage.** The pipeline already emits `sentence_confidence` via `make_report.py -word-confidence`. Flag segments with `mean_prob` < 0.6 for human review – this catches roughly 92% of NIV-N segments at modest false-positive cost.

3. **Do NOT promote entropy to a primary gate yet.** Marginal improvement over max-softmax, harder to explain to clients, doesn't beat hallucinations. Keep capturing it for archaeology.

4. **Schedule the all-20-beams decode change for next sprint** (Mission 6). The cheap top-3 preview is a near-clone of full entropy.

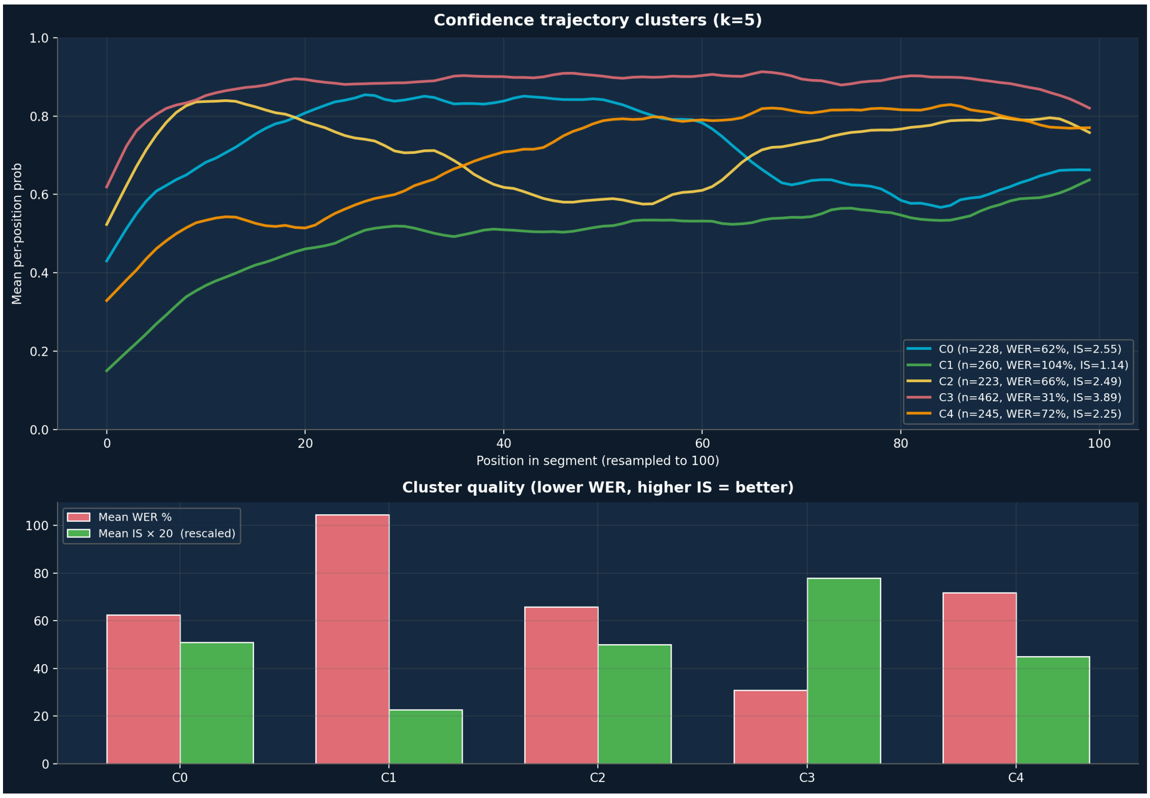

Figure 6: Top: $k = 5$ k-means cluster centroids on length-normalized per-position confidence trajectories. Bottom: per-cluster mean WER (coral) and mean IS rescaled by 20× (green).

Cluster	n	Mean WER %	Mean IS	NIV-Y rate	NIV-N rate
C3 (flat-high)	462	**30.8**	**3.89**	64.1%	3.0%
C0 (slow ramp-up)	228	62.5	2.55	10.1%	29.8%
C2 (flat-mid)	223	65.8	2.49	11.7%	33.2%
C4 (rising mid)	245	71.8	2.25	6.5%	40.8%
C1 (uncertain throughout)	260	**104.4**	**1.14**	0.0%	**91.9%**

5. **Trajectory clustering belongs in the runtime stack as a "give up early" signal** (Mission 11 candidate). Mid-decode trajectory shape predicts segment quality before the full sequence finishes.

10 Reproduce

```
python3 docs/_research-tools/generators/analyze_confidence_full.py \
    --confidence-sidecar /tmp/vsp_b3_1497_out/confidence-172610.json \
    --hypo               /tmp/vsp_b3_1497_out/hypo-172610.json \
    --baseline-csv       english_full_results/client_outputs/report/report.csv
```

Outputs land in `docs/confidence/confidence_full_analysis.md` and `presentation_materials_20260224`
The Markdown report is the canonical living version; this PDF is the stable snapshot.

Correlation	r
$r(full\_entropy, top3\_entropy)$	$+0.946$
$r(full\_entropy, WER)$	$+0.688$
$r(top3\_entropy, WER)$	$+0.699$
$r(full\_entropy, IS)$	-0.804
$r(top3\_entropy, IS)$	-0.803